$n = 25$

$n \longrightarrow n^{(n-1)/2}$

$2 \longrightarrow n/2$

$TC = O(n)$

$TC = O(n)$

Factors of any no always occur in pairs

$n = 20$

1, 2, 4, 5, 10, 20.

$n = 25$

1, 5, 25

$2 \longrightarrow \sqrt{n}$

$TC = O(\sqrt{n})$.

$n^{1/2}$

$n = 10^{8}$

$n^{1/2} = 10^{4}$

$\log n = 10$

All primes from 2 to n.

BF: TC: $O(n^{*}\sqrt{n}) = O(n^{3/2})$.

AS: $O(1)$.

d ↓ ✗ ✗ ↓

2 3 ✗ 5 ✗ 7 ✗ ✗ ✗

f

11 ✗ 13 ✗ ✗ ✗ 17 ✗ 19 ✗

AS: 26

✗ ✗ 23 ✗ ✗ ✗

$5^{*}5$

$\sqrt{5} \cdot \sqrt{5}$

$TC = O(n^{*}\log(\log(n)))$

$TC = O(\log n)$

$1 \cdots 1\ 0\ 1\ 1\ 0$

$2^{4}\ 2^{3}\ 2^{2}\ 2^{1}\ 2^{0}$

$1 \cdot 2^{4} + 0 \cdot 2^{3} + 1 \cdot 2^{2} + 1 \cdot 2^{1} + 0 \cdot 2^{0}$

$16 + 4 + 2 = 22.$

22		
2	11	0
2	5	1
2	2	1
2	1	0
2	0	1

10110

$TC = O(\log n)$

AND (&)

$$\begin{array}{r} 1\,0\,1\,0 \\ \&\ 0\,1\,1\,1 \\ \hline 0\,0\,1\,0 \end{array}$$

OR (|)

$$\begin{array}{r} 1\,0\,1\,0 \\ \text{OR}\ 0\,0\,1\,1 \\ \hline 1\,0\,1\,1 \end{array}$$

XOR (^)

$$\begin{array}{r} 1\,0\,1\,0 \\ \text{XOR}\ 0\,0\,1\,1 \\ \hline 1\,0\,0\,1 \end{array}$$

NOT (~)

$$\sim 1\,0\,1\,0$$
$$0\,1\,0\,1$$

Left Shift (<<)

$n = 1\,0\,1\,0 = 10$

$1\,0\,1\,0\,0 = 20$

→ Multiplies the num by 2.

Right shift (>>)

$n = 1\,0\,1\,0 = 10$

$1\,0\,1 = 5$

$n = 1\,0\,1\,1 = 11$

$1\,0\,1 = 5$

→ Divides the num by 2.

Set the k-th bit

1 0 1 1 0 1 (45)

$1 + 4 + 8 + 32 = 45$

$n = 45$

$n = 45,\ k = 1$

$n = 1\,0\,1\,1\,0\,1$

OR $0\,0\,0\,0\,1\,0$

$1\,0\,1\,1\,1\,1$

$n = 45,\ k = 1$

$n = 47.$

mask $= (1 << k) = 2^{k}$

$n = 1\,0\,1\,1\,0\,1,\ k = 5$

mask $= 1\,0\,0\,0\,0\,0$

$1\,0\,1\,1\,0\,1$

$0\,0\,0\,1$

$0\,0\,1\,0$

$0\,1\,0\,0$

$1\,0\,0\,0$

$1\,0\,0\,0\,0\,0$

$5 << k$

$0\,0\,0\,0\,0\,0\,0\,0\,0\,1\,0\,1$

Unset the k-th bit.

$n = 45$

$1\,0\,1\,1\,0\,1,\ k = 3$

OR: $1\,0\,0\,1\,0\,1$

$1\,0\,1\,1\,0\,1$

X XOR $0\,0\,1\,0\,0\,0$

$1\,0\,0\,1\,0\,1$

$1\,0\,1\,1\,0\,1$

X XOR $0\,0\,0\,0\,1\,0$

$1\,0\,1\,1\,1\,1$

$k = 1$

$1\,0\,1\,1\,0\,1,\ k = 3$

AND $1\,1\,0\,1\,1\,1$

$1\,0\,0\,1\,0\,1$

$(1 << k) = 0\,0\,1\,0\,0\,0$

$\sim(1 << k) = 1\,1\,0\,1\,1\,1$

Toggle.

$n = 45$

$1\,0\,1\,1\,0\,1,\ k = 3$

XOR $0\,0\,1\,0\,0\,0$

$1\,0\,0\,1\,0\,1$

XOR $1\,0\,0\,1\,0\,1$

$0\,0\,1\,0\,0\,0$

$1\,0\,1\,1\,0\,1$

XOR

$a \wedge a = 0$

$0 \wedge a = a$

$a \wedge a \wedge b = b$

$a \wedge b \wedge a = b$

$b \wedge a \wedge a = b$

$$\begin{array}{r} 0\,0\,0 \\ 1\,0\,1 \\ \hline 1\,0\,1 \end{array}$$

a	b	a^b
0	0	0
0	1	1
1	0	1
1	1	0